

Lab - WiFi security

Version 1.0, 31-10-2019, by Ivo Nijhuis

Introduction



Aircrack-ng, retrieved on 31-10-2019 from aircrack-ng.org

In this lab we'll have a closer look at WiFi security. We'll use a suite of tools called Aircrack-ng to assess the security of WiFi with WEP, WPA and WPA2 encryption.

To participate in this lab you'll need Virtual Box with the NE1 Debian VM running. You'll also need Wireshark. You probably already have these tools set up from the previous labs. If not: download & install!

Warning



Intercepting WiFi traffic, hacking passwords and decrypting encrypted messages are all potentially illegal activities. Only use these techniques on networks you own, or after first obtaining permission

Do NOT use Aircrack-ng tools on a network without first obtaining permission.

Installation

To install Aircrack-ng on your Debian Virtual Machine, enter:

```
sudo apt-get install aircrack-ng
```

It's that simple!

WiFi traffic capture

WiFi traffic travels through plain air, so in theory it's simple to capture. However, most WiFi adapters are configured in hardware to discard any frames that are not addressed to themselves. This is bad news for people that want to listen in, but good news for most people's privacy!

Specialised WiFi adapters are available for sale that do capture all traffic in the area, regardless of the destination. If we had such a customised adapter, we could use `airodump-ng` to record all WiFi traffic in the area, and save the captured frames to a capture file for analysis.

Since we do not have such dedicated hardware available, we have provided a number of sample capture files for you. These are available on your Virtual Machine in the directory:

```
/home/student/802.11_captures/
```

WEP password cracking

The oldest WiFi encryption standard is WEP. It has been proven to be cryptographically insecure, because it reuses certain codes (IV's or Initialisation Vectors). If a sufficient amount of traffic is captured, it is possible to recover the WiFi password by simply analysing the capture.

The file: `wep_64_ptw.cap` contains a capture of WiFi Traffic encrypted with WEP. To find out more about the recorded traffic, we can read the capture file into `airodump-ng`:

```
sudo airodump-ng -r wep_64_ptw.cap
```

1. What is the MAC address of the Access Point (Base Station)?
2. What is the encryption type used, according to the capture?

1. MAC address of AP:
 2. Encryption type:

Now, let's see if we can find what password was used, by analysing the capture. The WEP password will be shown as a *hexadecimal key*.

```
aircrack-ng wep_64_ptw.cap
```

3. What is the password for this WiFi network?

WPA password cracking

WPA and WPA2 are more secure than WEP. It is not possible to recover the password by simply analysing captured traffic. We can, however, try to brute force the password. This can be done by using a list of popular passwords, and just trying them one by one. If the WiFi network does not use a sufficiently strong password, we might be able to guess it after a number (or large number) of tries.

Such a list of passwords has been provided on your VM. It is called the RockYou password list.

See what's in the RockYou password list (*use space for the next page, or q to quit*):

```
more /usr/share/wordlists/rockyou.txt
```

If you see any password that you use personally, now is a good time to change it!

The file: `wpa.cap` contains a capture of WiFi Traffic encrypted with WPA. Read the capture file into `airodump-ng`:

```
sudo airodump-ng -r wpa.cap
```

4. What is the MAC address of the Access Point (Base Station)?
5. What is the SSID of the Access Point?
6. What is the MAC address of the station communicating with the Access Point?
7. What is the encryption type used, according to the capture?
8. What is the type of Cipher used, according to the capture?

4. MAC address of AP:
 5. SSID:
 6. MAC address of station:
 7. Encryption type:
 8. Cypher:

Now, let's see if we can find what password was used, by trying passwords from the RockYou password list:

```
aircrack-ng -w /usr/share/wordlists/rockyou.txt wpa.cap
```

9. What is the password for this WiFi network?

Now, let's try the same things for WPA2. You can find a capture in the file: `wpa2.cap`.

10. What is the MAC address of the Access Point (Base Station)?
11. What is the SSID of the Access Point?
12. What is the MAC address of the station communicating with the Access Point?
13. What is the encryption type used, according to the capture?
14. What is the type of Cipher used, according to the capture?
15. What is the password for this WiFi network?

10. MAC address of AP:
 11. SSID:
 12. MAC address of station:
 13. Encryption type:
 14. Cipher:
 15. Password:

Decrypting WiFi traffic

Using the password, we can decode encrypted WiFi traffic. A somewhat longer conversation is captured in the file: `wpa-psk-linksys.cap`

First, download this file to your laptop. The easiest way is probably to use an SFTP client like FileZilla, to connect to your VM. Check your VM's network settings: if your VM's network is set to NAT-network, you may want to change this back to NAT. With your network set to NAT, you can use SFTP to reach your VM at: `127.0.0.1:22`.

Open the downloaded file `wpa-psk-linksys.cap` in Wireshark.

16. Do you see any link layer (802.11) traffic?
17. Do you see any traffic from higher layer protocols like HTTP, TCP or IP?
18. Can you see IP addresses for the devices that are communicating?
19. What is the MAC address of the Access Point (Base Station)?
20. What is the MAC address of the station communicating with the Access Point?

- 16.
 - 17.
 - 18.
 19. MAC address of AP:
 20. MAC address of station:

Back on your VM, analyse the capture file, using `airodump-ng` and `aircrack-ng`.

21. Do you see the same or different MAC addresses for the Access Point (Base Station) and station as in Wireshark?
22. What is the SSID of the Access Point (it is also mentioned in the `probe` message)?
23. What is the password for this WiFi network?

- 21.
 22. SSID:
 23. Password:

Using the SSID of the Access Point, as well as the password we've recovered, we can now decrypt the capture file. Use:

```
airdecap-ng -e <SSID> -p <password> wpa-psk-linksys.cap
```

You will see that a decrypted capture file has been created:

```
wpa-psk-linksys-dec.cap
```

Transfer this capture file to your laptop and open it in Wireshark.



You should now be able to see IP addresses and also some different protocols like DNS, ICMP and ARP in action. Make a screenshot or picture of your capture in Wireshark to hand in.

Complete the lab

To wrap up the lab, make a single document containing your answers. Upload your document to DLO.